

Amutheezan Sivagnanam

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SUMMARY

- **Work Authorization Status:** US Person (US Green Card / Lawful Permanent Resident), Open to Relocation
- **PhD Candidate and Research Assistant** in Computer Science specializing in **machine learning**.
- **Six years** of academic research experience building **scalable reinforcement learning** and **deep learning** systems for efficient ML **training** and **deployment**. **Two years** of industrial experience in **software development**, **behavioral driven testing** and **CI/CD** while following **agile practices**.
- Good **publication** record with an **h-index of 6** and **98 citations**, featuring in top-tier **AI and ML** conferences.

EDUCATION

Ph.D. Computer Science, (August 2025): **Pennsylvania State University**, University Park, PA, Advisor: Dr. Aron Laszka

- **Dissertation:** Application of Deep Reinforcement Learning to Solve Optimization Problems in Transportation Domains

M.S. Computer Science, (August 2022): **University of Houston**, Houston, TX, Advisor: Dr. Aron Laszka

RESEARCH EXPERIENCE

Graduate Research Assistant, Pennsylvania State University, (August 2022 - Present)

- Devised and evaluated a **Transformer based multi-agent reinforcement learning algorithm** to solve the **emergency responder reallocation problem** that **reduces decision making time by 3 orders of magnitude**
- Leveraged a **deep reinforcement learning** approach to tackle the challenge of online vehicle routing with advance booking, enabling **prompt confirmation within seconds**.

Graduate Research Assistant, University of Houston, (September 2019 - August 2022)

- Introduced a novel **reinforcement learning** based solution approach to solve the problem of **online booking for offline vehicle routing problem** which resulted in **20 % reduction of operational costs** for the transit agency
- Introduced **novel problem formulation and algorithms** that **lower the energy costs by \$140k** annually for public transit agencies operating mixed fleets of buses.
- **Relevant Skills:** Python, C/ C++, Java, SQL, MySQL, MongoDB, PyTorch, TensorFlow, Scikit-learn, Transformers, Deep Learning, Reinforcement Learning, DQN, DDPG, TRPO, PPO, GRPO, RLHF, Bayesian Optimization, Clustering, Statistical Analysis, Data Mining, Ray, RLLib, OpenAI Gym, NumPy, pandas, matplotlib, CPLEX, Gurobi, Mosek, OR-Tools, Spark, Amazon Web Services, Git, Docker, GitHub, Linux, Bash, Object-Oriented Development, MATLAB

SELECT PUBLICATIONS

Total Citations 98 | H-Index 6 | Conference, Journal Publications 10

A. Sivagnanam et al.; Multi-Agent Reinforcement Learning with Hierarchical Coordination for Emergency Responder Stationing. International Conference on Machine Learning. **ICML 2024. Core Ranking A***

A. Sivagnanam et al.; Offline Vehicle Routing Problem with Online Bookings: A Novel Problem Formulation with Applications to Paratransit. International Joint Conference on Artificial Intelligence. **IJCAI 2022. Core Ranking A***, Invited Talk: Vienna Austria

A. Sivagnanam et al.; Minimizing Energy Use of Mixed-Fleet Public Transit for Fixed-Route Service. AAAI Conference on Artificial Intelligence. **AAAI 2021. Core Ranking A***

SOFTWARE ENGINEERING EXPERIENCE

Software Engineer, LSEG Technology (January 2018 - July 2019)

- Introduced **unit testing** for libraries in the Post-Trade C++ codebase. Implemented and validated **database changes** for Post-Trade products for the Singapore Stock Exchange using **Behavior Driven Development (BDD)** testing approaches in **Java**. Contributed to the **CI/CD pipeline** of the Post-Trade product using **Python and Git**.

Software Engineering Intern, WSO2 Lanka PVT Ltd (July 2016 - December 2016)

- Developed an **alert generation system** to monitor disease outbreaks by analyzing descriptive data, triggering email and SMS notifications for widespread incidents using **HL7/FHIR data**. Additionally, implemented a mechanism to assess hospital functionality, such as bed and oxygen cylinder availability, based on admission and discharge messages.
- **Relevant Skills:** Python, C++, JAVA, JS, PHP, SQL, Object Oriented Development, Github, Linux, Agile Practices, Behaviour Driven Testing, CI/CD

REFERENCES

Dr. Aron Laszka (PhD Supervisor): aql5923@psu.edu

Assistant Professor, Pennsylvania State University

<https://aronlaszka.com>

Dr. Abhishek Dubey: abhishek.dubey@vanderbilt.edu

Associate Professor, Vanderbilt University

<https://abhishekdubey.bio>

Dr. Weidong Shi: wshi3@uh.edu

Associate Professor, University of Houston